SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent
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COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Simulation-Based Assessment Question

1. **Develop an RL-based adaptive traffic signal control system and evaluate its performance.**

Ans: -

Aim

To develop a Reinforcement Learning-based adaptive traffic signal control system that learns to select the appropriate traffic signal phase according to the number of vehicles waiting on different roads, with the objective of reducing traffic congestion and vehicle waiting time.

Introduction

Traditional traffic signals generally operate using fixed timing. For example, a signal may remain green for 30 seconds regardless of whether there are many vehicles or only a few vehicles waiting. An adaptive traffic signal control system dynamically changes the signal based on current traffic conditions. In this simulation, Q-Learning, a model-free Reinforcement Learning algorithm, is used. The traffic signal controller acts as the RL agent. The number of waiting vehicles represents the state, and changing the traffic signal represents the action.

The agent receives a reward based on traffic congestion. If the number of waiting vehicles decreases, the agent receives a better reward. Through repeated interaction with the simulated traffic environment, the agent learns an effective traffic signal policy.

Reinforcement Learning Model
The system contains four important components:

Component	Description
Agent	Traffic signal controller
Environment	Road intersection
State	Number of waiting vehicles
Actions	North-South Green / East-West Green
Reward	Negative number of waiting vehicles

Q-Learning
Q-Learning maintains a **Q-table**.
The Q-table stores the expected value of taking an action in a particular state.
The Q-learning update equation is:

$$Q(s,a)=Q(s,a)+\alpha[r+\gamma \max Q(s',a')-Q(s,a)]$$

Where:

- s = current state
- a = current action
- r = reward
- s' = next state
- α = learning rate
- γ = discount factor

Algorithm-
Step 1- Initialize the Q-table with zeros.
Step 2- Generate an initial traffic condition.
Step 3- Observe the current state.
Step 4- Select an action using ε-greedy policy.
Step 5- Change the traffic signal according to the selected action.
Step 6- Allow some vehicles to pass.
Step 7- Generate new vehicles randomly.
Step 8- Calculate the reward.
Step 9- Update the Q-table using the Q-learning equation.
Step 10- Repeat the process for several episodes.
Step 11- After training, use the learned Q-table to control the traffic signal.
Step 12- Calculate the average waiting vehicles to evaluate performance.

Program

```
import numpy as np
import random
q = np.zeros((10, 10, 2))
alpha = 0.1
gamma = 0.9
epsilon = 0.2
for episode in range(500):
    ns = random.randint(0, 9)
    ew = random.randint(0, 9)
    for step in range(50):
        if random.random() < epsilon:
            action = random.randint(0, 1)
        else:
            action = np.argmax(q[ns, ew])
        if action == 0:
            ns = max(0, ns - random.randint(1, 3))
            ew = min(9, ew + random.randint(0, 2))
        else:
            ew = max(0, ew - random.randint(1, 3))
            ns = min(9, ns + random.randint(0, 2))
        reward = -(ns + ew)
        q[ns, ew, action] += alpha * (
            reward + gamma * np.max(q[ns, ew]) - q[ns, ew, action]
        )
print("Training completed")
ns, ew = 5, 5
total = 0
print("\nTraffic Signal Simulation")
for step in range(10):
    action = np.argmax(q[ns, ew])

    if action == 0:
        signal = "NORTH-SOUTH GREEN"
        ns = max(0, ns - random.randint(1, 3))
    else:
        signal = "EAST-WEST GREEN"
        ew = max(0, ew - random.randint(1, 3))
    waiting = ns + ew
    total += waiting
    print("Step", step + 1, "|", signal, "| Waiting Vehicles:", waiting)
print("\nAverage Waiting Vehicles:", total / 10)
```

Output-

```
Training completed

Traffic Signal Simulation
Step 1 | NORTH-SOUTH GREEN | Waiting Vehicles: 7
Step 2 | EAST-WEST GREEN | Waiting Vehicles: 4
Step 3 | NORTH-SOUTH GREEN | Waiting Vehicles: 2
Step 4 | EAST-WEST GREEN | Waiting Vehicles: 1
Step 5 | EAST-WEST GREEN | Waiting Vehicles: 0
Step 6 | NORTH-SOUTH GREEN | Waiting Vehicles: 0
Step 7 | NORTH-SOUTH GREEN | Waiting Vehicles: 0
Step 8 | NORTH-SOUTH GREEN | Waiting Vehicles: 0
Step 9 | NORTH-SOUTH GREEN | Waiting Vehicles: 0
Step 10 | NORTH-SOUTH GREEN | Waiting Vehicles: 0

Average Waiting Vehicles: 1.4
```

Result-

The RL-based adaptive traffic signal control system was successfully developed using Q-Learning. The agent learned to select either the North-South or East-West green phase based on the current traffic state. The performance was evaluated using the average number of waiting vehicles. The simulation demonstrates that Reinforcement Learning can be used to dynamically control traffic signals and reduce congestion.

Code of Conduct:

I U. Somesh Kumar (192425019) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
U. Somesh Kumar

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively